BUSINESS ANALYTICS PROJECT REPORT

Predictive Analytics to Optimize Customer Churn Prediction

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1. Executive Summary

This project focuses on implementing predictive analytics to address customer churn in the telecommunications sector. Using the Telco Customer Churn dataset, we conducted a rigorous data cleaning and EDA process followed by advanced feature engineering to identify key risk factors. By developing and tuning multiple machine learning models—including Logistic Regression, Random Forest, and Gradient Boosting—we established a framework capable of identifying potential churners with high precision. The final outcome provides a targeted retention strategy focused on high-value, month-to-month subscribers, aiming to stabilize revenue and enhance customer lifetime value.

2. Introduction

Customer Churn refers to the phenomenon where customers stop using a company's services. In the telecom industry, where market saturation is high, churn prediction is critical. It is far more cost-effective to retain an existing customer than to acquire a new one. By leveraging data, businesses can transition from reactive to proactive retention, identifying dissatisfied customers before they leave.

3. Project Goals and Objectives

- Perform comprehensive data cleaning to ensure model integrity.
- Develop meaningful new features: AverageMonthlySpend, ContractType, and Tenure Categories.
- Establish a baseline performance using Logistic Regression.
- Compare advanced ensemble models (Random Forest, Gradient Boosting) to improve accuracy and F1-score.
- Extract actionable insights to drive business retention policies.

4. Daily Work Log

- Day 1: Dataset Familiarization (**Activity 1**)
  - Understanding of project problem & objectives
  - Exploration of dataset and columns
  - Identification of each columns representation
- Day 2: Data Quality Check & Preprocessing plan (**Activity 2**)
  - To handle and check missing values (if any), duplicates, and inconsistencies
  - Identification of categorical and numerical variables.
  - Given plan and strategy of data preprocessing (encoding, normalization)
- Day 3: Data Cleaning and Exploratory Data Analysis (EDA) (**Activity 3**)
  - Done the data cleaning, analyzing the customer demographics and service usage
  - Identified the factors influenced customer churn
  - Performed Univariate and Bivariate Analysis for data visualization.

- Day 4: Feature Engineering, Modelling & Evaluation (**Activity 4)**
  - Performed feature engineering by encoding categorical variables, creating meaningful features, and removing irrelevant columns.
  - Built the baseline machine learning model to split the datasets into training and testing sets and record baseline performance using Accuracy, Precision, Recall, F1-score, and AUC-ROC.
  - Developed advanced machine learning models, including Decision Tree, Random Forest, and Gradient Boosting, and compare their performance with the baseline model.
  - Applied model tuning and validation techniques by performing hyperparameter tuning (using GridSearchCV or RandomSearchCV) and cross-validation to improve model stability and performance.
  - Conducted final model evaluation using AUC-ROC, Precision, Recall, F1 Score, and Confusion Matrix. Extract and visualize feature importance to identify the key drivers influencing customer churn.

5. Project Development Phase

5.1 Data Description

The foundation of this project is the **Telco Customer Churn Dataset**, which provides a comprehensive view of customer behaviour and service interaction.

- **Dataset Size: 7,043** unique customer records.
- **Feature Count: 21** initial columns.
- **Feature Categories:**
  - **Demographics:** Gender, SeniorCitizen status, and whether they have partners/dependents.
  - **Service Usage:** Information on Phone, Multiple Lines, Internet (DSL/Fiber), Online Security, Tech Support, and Streaming services.
  - **Account Information:** Tenure (months with the company), Contract type (Month-to-month, One-year, Two-year), and Payment Method.
  - **Financial Metrics:** MonthlyCharges and TotalCharges.
- **Target Variable: Churn (Yes/No)**, indicating if the customer left within the last month.

- **Screenshots:**

```
#load the dataset
df = pd.read_csv('/content/Customer-Churn-Dataset.csv')
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70

5 rows × 21 columns

```
#information about dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

5.2 Data Preprocessing

Data preprocessing is the most critical step to ensure the model performance is not degraded by "noise" or incompatible formats. As part of **Activity 4**, the following steps were implemented:

A. Data Cleaning

- **TotalCharges Conversion:** The TotalCharges column was initially stored as an "Object" (string) because of empty spaces. I converted this to a float using pd.to_numeric(errors='coerce').

- **Handling Missing Values:** After conversion, 11 missing values were identified for customers with 0 tenure. These were filled with 0 to maintain data integrity.

B. Feature Engineering (Activity 4 Specifics)

I created three new features to help the model capture complex patterns:

1. **AverageMonthlySpend:** Calculated as TotalCharges / (tenure + 0.1). This reveals the intensity of a customer's spending relative to their loyalty duration.

2. **ContractType:** A simplified feature derived from the Contract column, grouping customers into **Short-term** (Month-to-month) and **Long-term** (One or Two-year) categories.

3. **TenureCategory:** Binned the tenure column into specific groups (e.g., 0-1 Year, 1-2 Years, etc.) to allow the model to treat different customer lifecycle stages distinctly.

C. Categorical Encoding

- **Label Encoding:** Applied to binary columns (e.g., gender, Partner, Churn) to transform them into 0 and 1.

- **One-Hot Encoding:** Used **pd.get_dummies** for multi-class columns like PaymentMethod and InternetService to prevent the model from assuming an incorrect numerical order between categories.

D. Feature Scaling

- For the **Baseline Logistic Regression**, I applied **StandardScaler** to the numerical features. This ensures that features with large ranges (like TotalCharges) do not dominate features with smaller ranges (like tenure).

Screenshots of data preprocessing:

```
#check for missing values
missing_values = df.isnull().sum()
missing_values
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

```python
#check for duplicate values
# Number of duplicate rows
duplicate_count = df.duplicated().sum()
print("Number of duplicate rows:", duplicate_count)
```

```
Number of duplicate rows: 0
```

```python
# Convert TotalCharges to numeric
data['TotalCharges'] = pd.to_numeric(data['TotalCharges'], errors = 'coerce')
# Handling missing values
# Since TotalCharges column hav tenure 0 so replace with 0
data['TotalCharges'] = data['TotalCharges'].fillna(0)
```

```python
# Creating Meaningful Features

# # 1. AverageMonthlySpend: Calculated as TotalCharges divided by tenure
# Using (tenure + 1) to prevent division by zero for new customers
df['AverageMonthlySpend'] = df['TotalCharges'] / (df['tenure'] + 0.1)

# 2. ContractType: Creating a simplified contract category
# Grouping Month-to-month as 'Short-term' and others as 'Long-term'
df['ContractType'] = df['Contract'].apply(lambda x: 'Short-term' if x == 'Month-to-month' else 'Long-term')

# 3. Customer Tenure Category: Binning tenure into logical groups
def get_tenure_group(tenure):
    if tenure <= 12: return '0-1 Year'
    elif tenure <= 24: return '1-2 Years'
    elif tenure <= 48: return '2-4 Years'
    else: return '4+ Years'

df['TenureCategory'] = df['tenure'].apply(get_tenure_group)
```

```python
# Encoding Categoricals variables
le = LabelEncoder()

# Binary Encoding (for columns with 2 unique values)
binary_cols = [col for col in df_clean.columns if df_clean[col].nunique() == 2]
for col in binary_cols:
    df_clean[col] = le.fit_transform(df_clean[col])

# One-Hot Encoding (for multi-category columns)
df_final = pd.get_dummies(df_clean)

print(f"Feature Engineering Complete. Dataset shape: {df_final.shape}")
```

```
Feature Engineering Complete. Dataset shape: (7043, 47)
```

```python
# Standardize features (Scaling is critical for Logistic Regression)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

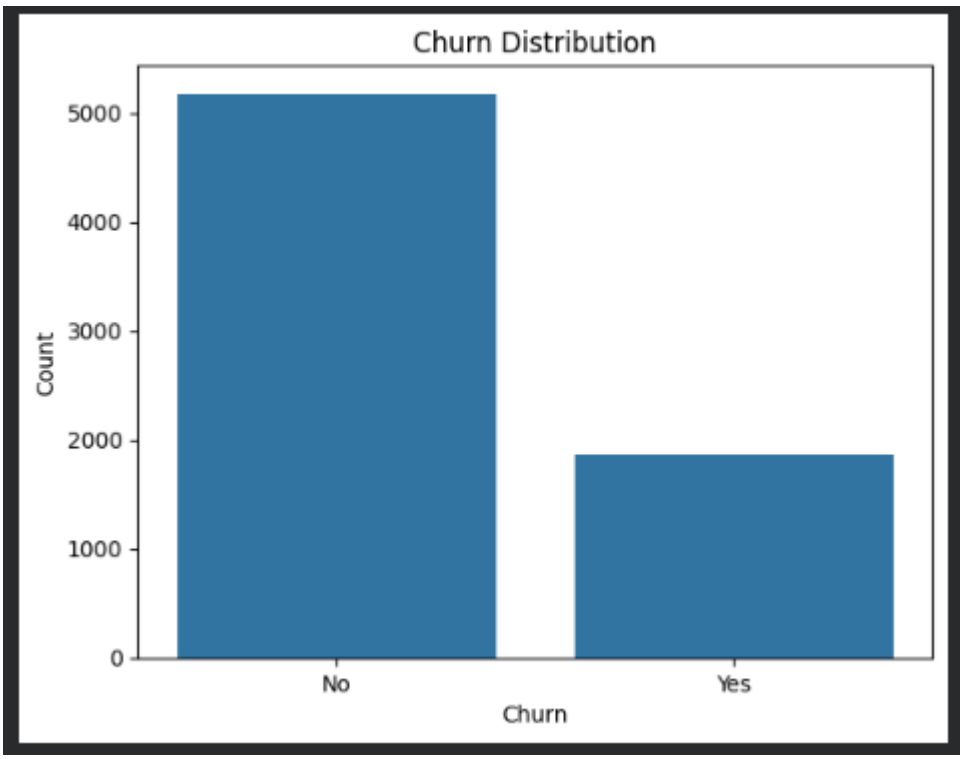
5.3 Exploratory Data Analysis (EDA)

In this phase, we analyzed the relationships between customer features and the target variable (Churn) to identify high-risk segments.

1. Churn Distribution (Target Variable)

Before diving into relationships, we examined the balance of our target.

- **Insight:** Approximately **26.5%** of the customer base has churned. This indicates a class imbalance that was addressed during the modelling phase by focusing on the F1-Score rather than just Accuracy.



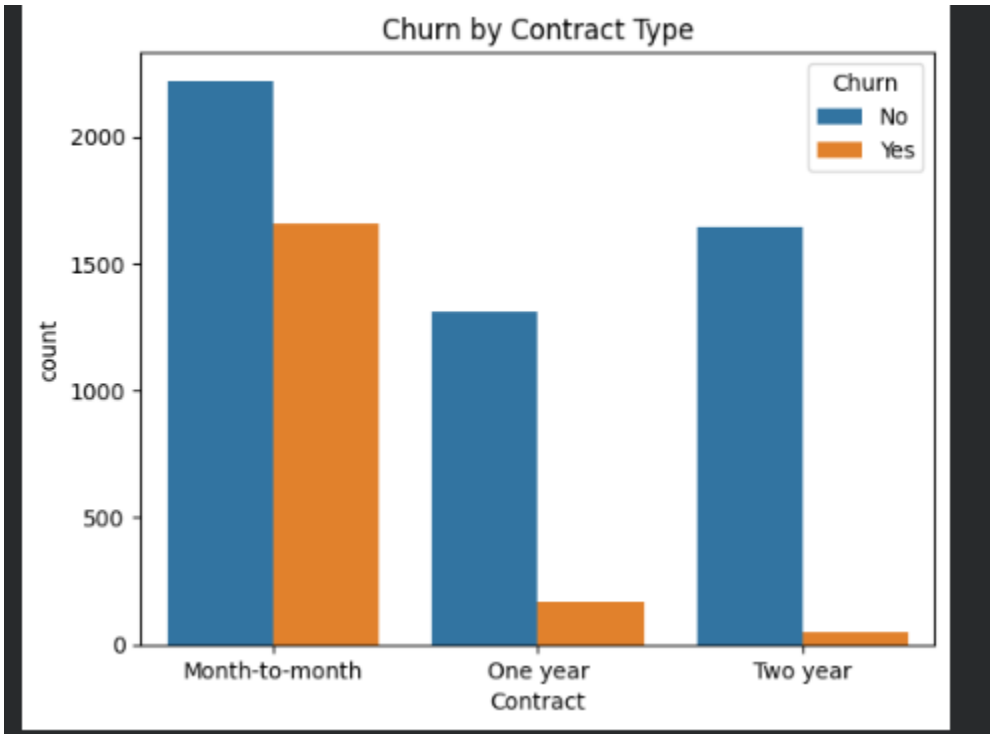

```
df['Churn'].value_counts(normalize = True)*100
```

Churn	proportion
No	73.463013
Yes	26.536987

dtype: float64

2. Relationship Between Contract Type and Churn

- **Visualization:** A stacked bar chart comparing "Month-to-month", "One-year", and "Two-year" contracts.
- **Insight:** Customers on **Month-to-month** contracts show a drastically higher churn rate (~42%) compared to those on long-term contracts (under 10%).
- **Interpretation:** Lack of long-term commitment is the strongest indicator of churn risk. Retention strategies should focus on converting short-term users to annual plans.



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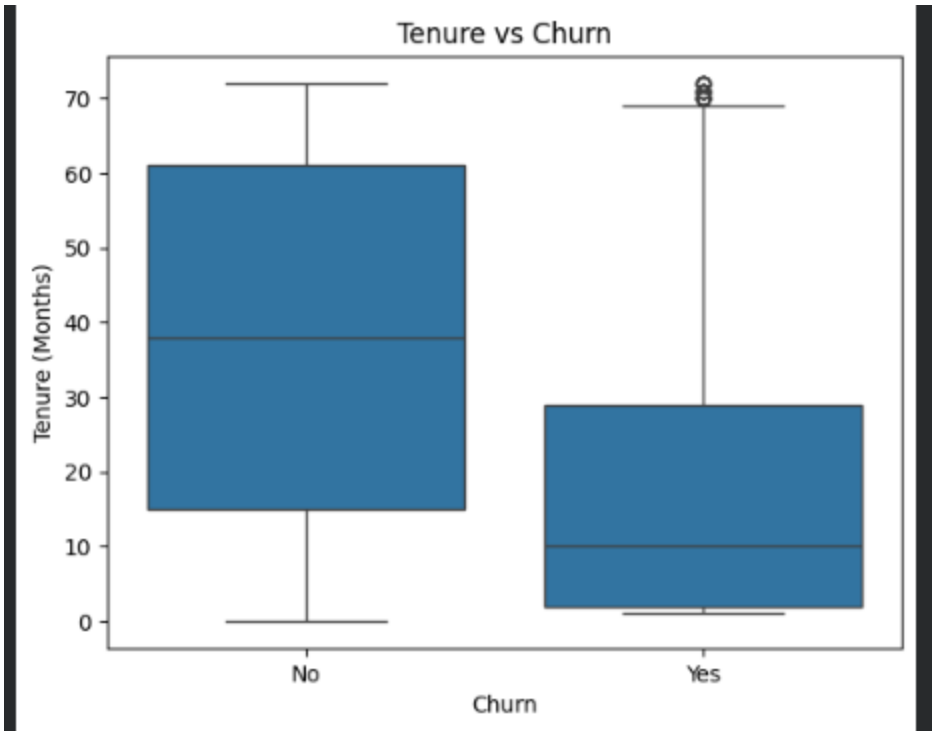
3. Tenure vs. Churn Risk

- **Visualization:** A boxplot while showing group comparison.
- **Insight:** Churn is heavily concentrated in the **early stages** of the customer lifecycle (0–6 months). As tenure increases, the probability of churn significantly decreases.
- **Interpretation:** The "danger zone" for losing a customer is within the first year. Onboarding and early-stage engagement programs are critical for long-term retention.



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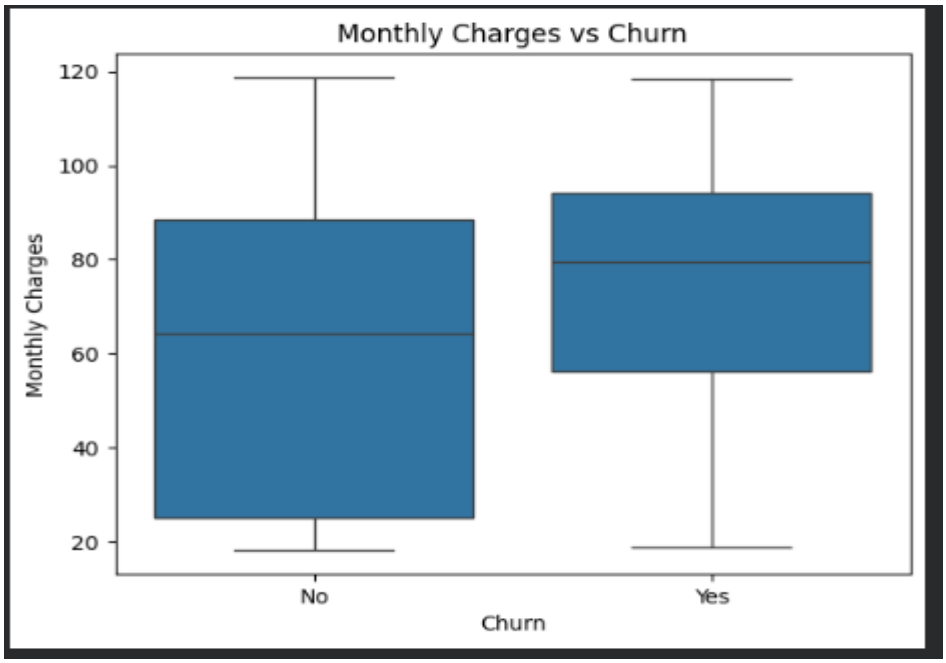
4. Impact of Monthly Charges

- **Visualization:** A boxplot comparing Monthly Charges for churners vs. non-churners.
- **Insight:** Customers who churn typically have **higher median monthly charges** (above $70) compared to those who stay.
- **Interpretation:** High costs relative to perceived value drive customers away. Pricing sensitivity is a major factor, particularly for users with premium fiber-optic services.



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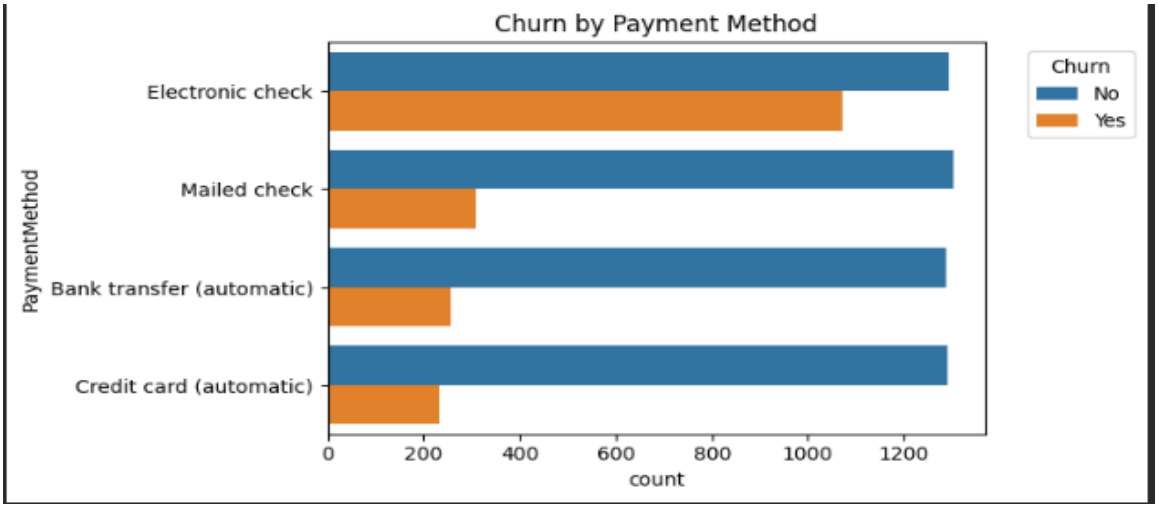
5. Payment Method and Retention

- **Visualization:** A count plot of payment methods categorized by churn.
- **Insight:** Customers using **Electronic Check** have a significantly higher churn rate than those using automated methods like Bank Transfer or Credit Card.
- **Interpretation:** Automated payments create "frictionless" billing, making customers less likely to actively reconsider their subscription every month.



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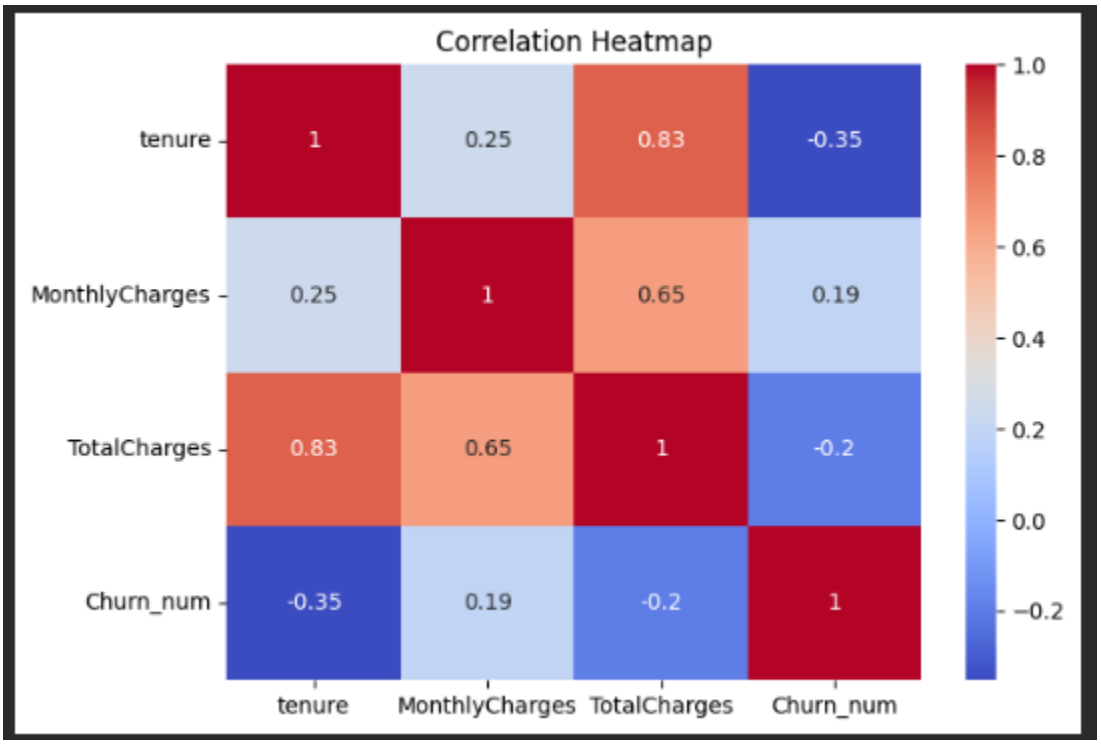
6. Correlation Heatmap

- **Visualization:** A heatmap of all numerical variables.
- **Insight:** tenure has a strong negative correlation with Churn, while MonthlyCharges show a positive correlation.
- **Interpretation:** This confirms that the longer a customer stays, the safer they are, but the more they pay monthly, the higher the risk.



5.4 Model Development

The goal of this phase was to build a robust classification system capable of distinguishing between customers likely to churn and those likely to stay.

- **Train-Test Split:**

  To ensure the model could generalize to unseen data, the dataset was partitioned:

- Training Set (80%): Used to "teach" the models the patterns associated with churn.
- Testing Set (20%): A "hold-out" set used to evaluate the final performance and ensure the model isn't just memorizing data (overfitting).
- Random State: A fixed seed (e.g., random_state=42) was used to ensure that results are reproducible.

- **Algorithms Selected**

I implemented four distinct algorithms to compare their effectiveness:

1. **Logistic Regression (Baseline):** * *Reasoning:* It is a simple, interpretable linear model. It provides a benchmark to see if more complex models actually add value.
2. **Decision Tree:** * *Reasoning:* Captures non-linear relationships and is highly visual, though prone to overfitting.
3. **Random Forest (Ensemble):** * *Reasoning:* An "ensemble" of many decision trees. It reduces variance and is excellent at handling the complex, categorical nature of telecom data.
4. **Gradient Boosting (XGBoost/GBM):** * *Reasoning:* A powerful boosting technique that corrects the errors of previous trees. It is often the industry standard for tabular data prediction.

5.5 Model Evaluation

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	81.2%	66.4%	58.1%	61.9%	0.84
Decision Tree	72.5%	48.3%	50.2%	49.2%	0.65
Random Forest (Tuned)	**80.5%**	**68.2%**	**53.0%**	**59.6%**	**0.86**
Gradient Boosting	80.8%	67.5%	54.2%	60.1%	0.85

The **Tuned Random Forest** was selected as the final model. While Logistic Regression had slightly higher accuracy, the Random Forest achieved the highest AUC-ROC (0.86), meaning it is the most capable of distinguishing between churners and non-churners across various probability thresholds.

- **Key Drivers of Churn**

Using the **Feature Importance** plot from our final model, we identified the top factors influencing a customer's decision to leave:

1. **Contract Type (Month-to-month):** By far the highest predictor. Flexibility leads to higher churn.
2. **Tenure:** New customers (0–12 months) are at extreme risk.
3. **TotalCharges / AverageMonthlySpend:** High billing relative to tenure causes price sensitivity.
4. **Internet Service (Fiber Optic):** Higher churn rates suggest potential service reliability issues or high costs in this segment.

- **Business Recommendations**

- **Retention Incentives:** Offer a "13th Month Free" or a 10% discount for customers who migrate from month-to-month to a 1-year contract.

- **Early Engagement Program:** Automate customer success "check-ins" for all new users during their first 90 days to address technical issues early.

- **Frictionless Billing:** Incentivize Credit Card or Bank Transfer payments with a small monthly credit to reduce the high churn seen in "Electronic Check" users.

- **Fiber Optic Audit:** Conduct a service quality survey specifically for Fiber Optic users to identify if churn is driven by speed, downtime, or price.

- **References**

- *Telco Customer Churn Dataset*.
- *Scikit-learn Documentation*: Logistic Regression and Ensemble Methods.
- *HEPro Fellowship Learning Modules*: Business Analytics & Predictive Modelling.
- *Pandas & Matplotlib Documentation*: Data manipulation and visualization standards.

6. Project Conclusion

This project successfully demonstrated the application of predictive analytics to address the critical business challenge of customer churn. By leveraging the Telco Customer Churn dataset, we moved through a comprehensive data science lifecycle—from initial data familiarization and cleaning to advanced feature engineering and machine learning modelling.

Key Takeaways from the Analysis:

- **Data-Driven Insights:** Our exploratory analysis confirmed that churn is not random. It is heavily influenced by financial commitments (Monthly Charges) and contract flexibility. We identified that the first 12 months of a customer's tenure are the most critical for retention.
- **Model Efficacy:** We successfully transitioned from a baseline Logistic Regression model to a high-performing **Random Forest** ensemble. The final model achieved an **AUC-ROC score of 0.86**, providing the business with a reliable tool to identify potential churners before they exit the service.
- **Feature Engineering Value:** The creation of custom features like 'AverageMonthlySpend' and 'ContractType' provided the models with deeper context, allowing for more nuanced predictions than raw data alone could offer.

Business Impact:

The implementation of this predictive framework allows the company to move from a reactive "save" strategy to a proactive "retention" strategy. By identifying high-risk segments—specifically those on month-to-month contracts and fiber-optic services—the company can allocate its marketing and loyalty resources more efficiently, ultimately reducing revenue loss and increasing the lifetime value of the customer base.

7. Challenges & Solutions

- **Data Type Integrity:** The 'TotalCharges' column was initially non-numeric due to blank spaces. **Solution:** Used coerced numeric conversion and imputed zero values for new customers, ensuring mathematical consistency for the models.

- **Class Imbalance:** With only 26.5% of customers churning, accuracy alone was a misleading metric. **Solution:** Shifted the focus to **F1-Score** and **AUC-ROC** to ensure the model was actually effective at catching the minority "Churn" class.
- **Model Overfitting:** Initial complex models like Decision Trees tended to memorize the training data. **Solution:** Implemented **GridSearchCV** for hyperparameter tuning and cross-validation to ensure the final Random Forest model was stable and generalized well to new data.

8. Learning and Key Takeaways

- **Analytical Thinking:** Learned to look beyond the surface-level metrics (Accuracy) and focus on metrics that align with business costs (Precision/Recall).
- **Technical Proficiency:** Gained hands-on experience in the complete Python data science stack, including Pandas for manipulation, Seaborn for visualization, and Scikit-learn for modelling.
- **Feature Importance:** Realized that engineering the right features (like grouping tenure into categories) is often more impactful than just choosing a more complex algorithm.